BENCHMARK BRIEF

Box3D Metal M4 Pro Performance

End-to-end crossover evidence, interpretation, and the limits of the recorded Apple M4 Pro benchmark results.

APPLE SILICON / METAL COMPUTE / BOX3D OVERLAY

Measured platform

Component	Recorded value
Machine	MacBook Pro Mac16,8
SoC	Apple M4 Pro - 12 CPU / 16 GPU cores
Memory	24 GB unified
OS / Metal	macOS 26.7 / Metal 4 compiler 32023.883
Compiler	AppleClang 21.0.0 Release, FP contraction off
World settings	8 workers, 4 substeps

Whole-world timings include preparation, submission, synchronization, finalization, and readback. The fused primitive is explicitly labeled because it excludes full-world bookkeeping.

Results

Workload	Crossover	Best/large result	Bodies
Position primitive	~131K	1.278x	524,288
Fused 4-substep primitive	~2K	10.584x	524,288
Unconstrained whole world	largest point	1.146x	524,288
Convex contacts	~262K	1.098x	262,144
Mesh contacts	~8K	1.646x	131,072
Distance joints	~131K	1.158x	524,288
Parallel joints	none stable	0.974x	1,048,576
Experimental GPU finalization	none	0.79x	524,288

What the data says

- Command fusion is the largest demonstrated architectural win.
- Experimental finalization arithmetic is correct but 27% slower at the paired 524,288-body median.
- Mesh contacts cross earlier than convex-wide stacks in the recorded workloads.
- Distance joints benefit only at very large scale.
- Parallel joints expand compatibility but do not justify default GPU routing.

No universal threshold

GPU frequency, worker scheduling, topology, contact density, and CPU-resident stages move crossover. The API therefore requires a caller-selected awake-body threshold.

```
./scripts/run-benchmarks.sh ../box3d-metal-worktree
```

Run each complete executable in at least three separate processes and compare medians. Do not use GPU kernel time alone as a whole-world claim.

Next performance work

- Move shape finalization and AABB generation into the existing command graph.
- Retain body and supported-joint state across world steps.
- Add joint types only with mode matrices and whole-world evidence.
- Implement GPU pair generation and broad phase after bounds ownership is resolved.